Quiz 8 — Solutions

M 375T Mathematical Foundations of Machine Learning — Fall 2026

The best line through four points (10 points)

Step 1: the c_i for a fixed q. For each i, $\|x_i - c_i q\|^2 = \|x_i\|^2 - 2c_i\,(q \cdot x_i) + c_i^2$ (using $\|q\| = 1$), an upward parabola in c_i with minimum at $c_i = q \cdot x_i$, where it equals $\|x_i\|^2 - (q \cdot x_i)^2$. [3 pts]

Step 2: the q. Summing,

$$\frac{1}{4}\sum_i \|x_i - c_i q\|^2 = \frac{1}{4}\sum_i \|x_i\|^2 - \frac{1}{4}\sum_i (q \cdot x_i)^2 = \frac{1}{4}\sum_i \|x_i\|^2 - q^{\mathsf{T}} S q, \qquad S = \frac{1}{4}\sum_i x_i x_i^{\mathsf{T}}.$$

So q must maximize $q^{\mathsf{T}} S q$ over unit vectors: q is the top eigenvector of S and the maximum is λ_1. [3 pts]

Step 3: the numbers. $x_1 x_1^{\mathsf{T}} = x_2 x_2^{\mathsf{T}} = \begin{pmatrix} 9 & 9 \\ 9 & 9 \end{pmatrix}$, $x_3 x_3^{\mathsf{T}} = x_4 x_4^{\mathsf{T}} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, so $S = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$. $\det(S - \lambda I) = (5-\lambda)^2 - 16 = 0 \Rightarrow \lambda_1 = 9,\ \lambda_2 = 1$; $(S - 9I)q = 0$ gives $q = (1,1)/\sqrt{2}$ (or its negative). [2 pts]

Answer. $q = (1,1)/\sqrt{2}$ and $c_i = q \cdot x_i$: $c_1 = 3\sqrt{2}$, $c_2 = -3\sqrt{2}$, $c_3 = c_4 = 0$ (signs flip together if q is replaced by $-q$). Check, not required: the minimum is $\frac{1}{4}(18 + 18 + 2 + 2) - 9 = 10 - 9 = 1 = \lambda_2$; the residuals $x_i - c_i q$ are $(0,0), (0,0), (1,-1), (-1,1)$, squared lengths $0, 0, 2, 2$, mean 1. [2 pts]

Scoring: 3 for $c_i = q \cdot x_i$ (completing the square or $\partial/\partial c_i = 0$), 3 for reducing to $\max q^{\mathsf{T}} S q =$ top eigenvector, 2 for S and its eigenpair, 2 for q and the four c_i. A student who guesses $q = (1,1)/\sqrt{2}$ from the picture and computes the c_i, but gives no argument that this is the minimizer, gets at most 4. Same form as Practice 8, B2.